Brainstorming & Idea Prioritization Template

Date	05 July 2026
Team ID	AIML-HDI-2026
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	3 Marks

Team: Nimmakayala Venkata Satheesh Kumar (Team Lead), Cherukuri Sailaja Kiranmai (Group Member).

Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Nimmakayala Venkata Satheesh Kumar	Use a dual-branch hybrid neural network that separates core UNDP indicators (life expectancy, schooling, GNI) from auxiliary indicators, since the two groups contribute differently to the HDI formula.	Machine Learning Model	1
2	Nimmakayala Venkata Satheesh Kumar	Train with AdamW optimizer and MSE loss over many epochs to get a smooth regression output, then derive tier classification via threshold rules instead of a separate classifier.	Machine Learning Model	1
3	Cherukuri Sailaja Kiranmai	Source the dataset from the official Kaggle Human Development Index (Full) dataset and profile it for missing values and column-naming inconsistencies across years.	Data Processing	2
4	Cherukuri Sailaja Kiranmai	Use GPU-accelerated cuDF for missing-value imputation so preprocessing scales to multi-year datasets without slowing down the Kaggle notebook.	Data Processing	2
5	Nimmakayala Venkata Satheesh Kumar	Deploy the trained model behind a lightweight Flask backend with an HTML/CSS dashboard, tunnelled live via LocalTunnel directly from the Kaggle notebook session.	Backend / Deployment	3
6	Cherukuri Sailaja Kiranmai	Maintain a GitHub repository documenting every phase (ideation through demonstration) and monitor the live web app after each deployment.	Documentation / DevOps	3

Step 2: Idea Prioritization

Rate each group idea on feasibility and importance, then select the final idea(s) to move forward with.

Group No.	Final Idea	Feasibility	Importance	Priority	Selected
1	Dual-branch hybrid neural network (primary + secondary indicator branches) trained with AdamW/MSE, with tier classification derived from the regression output.	High	High	1	Yes
2	GPU-accelerated (cuDF) data cleaning and dynamic column mapping to handle dataset naming variation across years.	High	High	2	Yes
3	Flask + HTML dashboard tunnelled live via LocalTunnel, with GitHub-tracked documentation across all phases.	High	Medium	3	Yes